Signals & Systems (151-0575-01) Exam, February 6th, 2020 - Scoring rubric (compact, with figures)

Condensed from the official corrected solution. Each criterion heading gives the label, name and maximum. Context is stated once per problem; 'Solution' is the official reference answer; 'Grading' gives the official point allocation for partial credit. Multiple-choice/true-false items: the key is given as option letter and text; all-or-nothing, no justification required. 67 points in total.

Problem 1 context: Consider the causal continuous-time system dy/dt(t) = a y(t) + u(t), with transfer function H(s) = 1 / (s - a). Assume u(t) is piece-wise constant over intervals Ts: u(t) = u[n] for n Ts <= t < (n+1) Ts. The system is discretized such that y[n] = (1 + a Ts) y[n-1] + Ts u[n-1], where y[n] = y(n Ts) and u[n] = u(n Ts).

1.1.a - Discretization method - 2 points:

Question: Which discretization method is used to obtain the discrete-time system?

Solution: Forward Euler discretization.

1.2.a - z as a function of s - 3 points:

Question: Based on the continuous-time system and its discretized version, determine the z-transform variable as a function of s, i.e. a relation of the form z = f(s). Justify your answer.

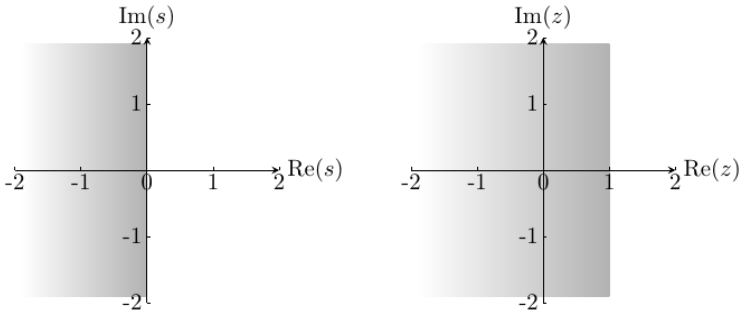
Solution: Z-transform of the DT system: Y(z) = (1 + a Ts) z^-1 Y(z) + Ts z^-1 U(z), so H(z) = 1 / (z/Ts - (1 + a Ts)/Ts). Matching denominators with H(s): z/Ts - (1 + a Ts)/Ts = s - a => z = 1 + s Ts (equivalently, z = e^(s Ts) ~ 1 + s Ts, the forward Euler approximation).

1.3.a - Stability region mapping - 3 points:

Question: The region of stability in the continuous domain is Re(s) < 0. Using the relation found in the previous part, describe the region of the z-plane to which stable continuous-time poles are mapped, and explain how you obtain it.

Solution: With z = 1 + s Ts, Re(s) < 0 maps to Re((z - 1)/Ts) < 0, i.e. Re(z) < 1: the open half-plane to the left of the vertical line Re(z) = 1. Rubric: 1 point for the correct region, 2 points for the correct argument.

Grading: 1 point for the correct region, 2 points for a correct argument.

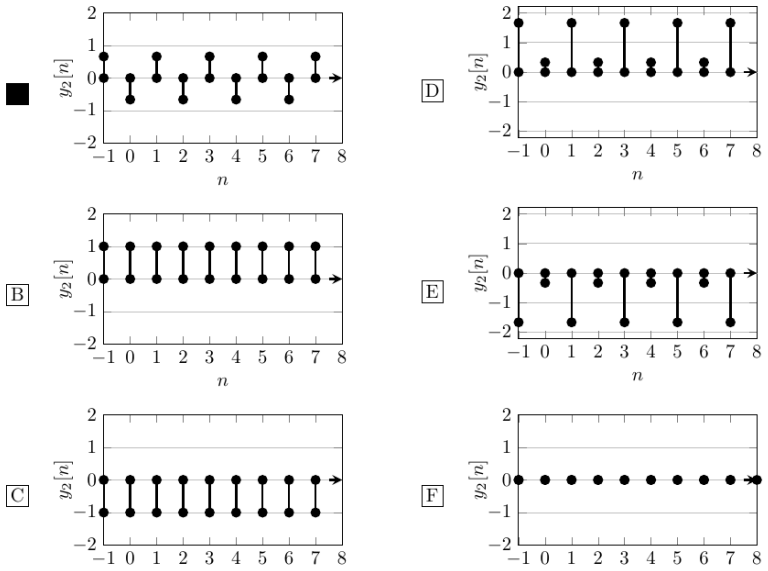


Official solution figure: stable s-region (left) and its image Re(z) < 1 (right, shaded).

2.a - High-pass output plot - 2 points:

Question: Consider the signal u[n] = 1 + cos(pi n) for all n, processed by the system y_1[n] = y_1[n-1] + 0.5 (u[n] - y_1[n-1]), y_2[n] = u[n] - y_1[n]. Which of six plots (Plot A to Plot F, each showing a candidate y_2[n] for n = -1..8) is the output y_2[n]? (see figure below) No explanation is needed.

Key: option A - Plot A



Answer options as printed; the filled box marks the correct option (A).

Problem 3 context: Consider a causal, unstable LTI system G with transfer function H(z) = b / (1 + a z^(-1)), which is stabilized with a negative feedback loop using a constant gain K = 3: the error r[n] - y[n] is multiplied by K to give u[n], which is the input of G, whose output is y[n].

3.1.a - Identify a and b - 6 points:

Question: To identify the coefficients of H(z), the signals r_1[n] = 1 and r_2[n] = (-1)^n for all n are given as reference to the closed-loop system. The measured outputs are y_1[n] = 1/2 and y_2[n] = 3/2 (-1)^n, respectively. Calculate the values of a and b. Justify your answer.

Solution: Closed loop: W(z) = K H(z) / (1 + K H(z)) = K b / (a z^-1 + (1 + K b)) (2 points). W(1) = 3b / (a + 1 + 3b) = 1/2 and W(-1) = 3b / (-a + 1 + 3b) = 3/2, i.e. 6b = a + 1 + 3b and 2b = -a + 1 + 3b. Summing: 8b = 2 + 6b => b = 1, then a = 2 (2 points each for setup and solution).

Grading: 2 points for the closed-loop transfer function W(z); 2 points for the system of equations; 2 points for a = 2, b = 1.

3.2.a - Closed-loop stability range - 2 points:

Question: Given b = 2, for which values of a is the closed-loop system BIBO stable? Justify your answer.

Solution: Closed-loop denominator a z^-1 + (1 + 6) = a z^-1 + 7 gives the pole z = -a/7 (magnitude |a|/7). BIBO stable iff |a/7| < 1, i.e. |a| < 7.

4.a - Filter design - 10 points:

Question: Consider the signal u[n] = a_0 + a_1 cos(pi/2 n) + a_2 (-1)^n, n = ..., -1, 0, 1, ..., where a_0, a_1, a_2 in R are fixed but unknown constants. Design a causal, stable, LTI filter G so that if u has the form above and y = G u, then y[n] = a_0 + 2 a_2 (-1)^n for all n. Justify your answer.

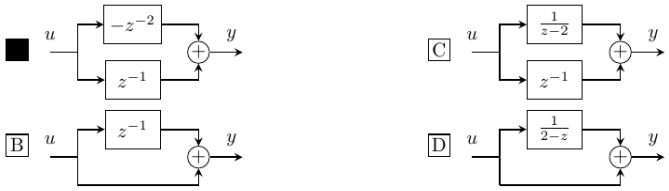
Solution: Write u[n] = a_0 e^(j0n) + a_1 (e^(j pi/2 n) + e^(-j pi/2 n))/2 + a_2 e^(j pi n) (1 point). Need H(1) = 1, H(-1) = 2, H(e^(j pi/2)) = 0, H(e^(-j pi/2)) = 0 (1 point each). A FIR filter is guaranteed stable (1 point); causality by multiplying by z^-N (1 point). Zeros at +-j: H_1(z) = alpha (z^2 + 1), with alpha = 1/2 so H_1(1) = H_1(-1) = 1. A first-order factor H_2(z) = 3/2 z - 1/2 adjusts the gains at Omega = 0 and Omega = pi (2 points, one per condition). Result: H(z) = 1/2 (z^2 + 1)(3/2 z - 1/2) z^-3, where z^-3 makes the filter causal (1 point); check: H(1) = 1, H(-1) = 2, H(+-j) = 0. Any causal stable filter meeting the four frequency conditions is acceptable.

Grading: 1 point for the frequency decomposition; 1 point for each of the 4 conditions on H; 1 point for stability (FIR); 1 point for causality; 2 points for the gain-adjusting factor; 1 point for the final causal H(z). Any causal, stable filter meeting the four conditions is acceptable.

5.a - Block diagram choice - 2 points:

Question: Let {u[n]} = {n s[n-1]} be the input sequence applied to a causal LTI system G (s[n] is the unit step). Which of the following block diagrams could represent the transfer function of G if the output is {y[n]} = {s[n-2]}? No explanation is needed.

Key: option A - u feeds two parallel blocks, -z^(-2) and z^(-1), whose outputs are summed to give y



Answer options as printed; the filled box marks the correct option (A).

Problem 6 context: Consider the transfer function H(z) = 0.25 (z + 2 + z^(-1)).

6.1.a - Frequency response - 2 points:

Question: Describe the magnitude and phase response of H(z) for Omega in [0, pi] (give H(Omega) explicitly).

Solution: H(Omega) = 0.25 (e^(j Omega) + 2 + e^(-j Omega)) = 1/2 (1 + cos(Omega)): real and non-negative, so |H| falls from 1 at Omega = 0 to 0 at Omega = pi (|H(pi/2)| = 1/2), and the phase is 0.

6.2.a - FIR/IIR and causality - 2 points:

Question: Determine if the system is FIR or IIR, causal or non-causal.

Solution: FIR (moving-average type) and non-causal (because of the z term). 1 point per condition.

Grading: 1 point per property (FIR; non-causal).

6.3.a - Output for cosine input - 2 points:

Question: Given the input signal u[n] = cos(pi/3 n), determine the output y[n].

Solution: H(pi/3) = 1/2 (1 + cos(pi/3)) = 3/4 with zero phase, so y[n] = 3/4 cos(pi/3 n).

Problem 7 context: Consider a causal LTI system G for which, given the input {u[n]} = {s[n]} - {s[n-2]} (s[n] is the unit step), the corresponding output is {y[n]} = G{u[n]} = {delta[n-1]} + {delta[n-4]}.

7.1.a - Impulse response - 6 points:

Question: Determine the impulse response sequence {h[n]}. Justify your answer.

Solution: s[n] - s[n-2] = delta[n] + delta[n-1], so h[n] + h[n-1] = delta[n-1] + delta[n-4]. With causality, recursing h[n] = -h[n-1] + delta[n-1] + delta[n-4] gives h[n] = {0, 1, -1, 1, 0, 0, ...} for n >= 0 (h[n] = 0 for n < 0). Rubric: 4 points for a convincing procedure, 2 for the correct sequence.

Grading: 4 points for a convincing procedure for obtaining h[n]; 2 points for the correct sequence.

7.2.a - BIBO stability - 1 points:

Question: Determine whether system G is BIBO stable.

Solution: sum |h[n]| = 3 < infinity, so the system is BIBO stable.

Problem 8 context: Consider an LTI system G, with G != 0, with frequency response H(Omega) = |H(Omega)| e^(j Theta_H(Omega)). State whether the following statement is guaranteed to be true. No explanation is needed.

8.1.a - True/false 1 - 1 points:

Question: If u is a complex exponential sequence with frequency Omega, then y = G u has the same frequency as u.

Key: option A - True

8.2.a - True/false 2 - 1 points:

Question: If u[n] = cos(Omega_1 n) + cos(Omega_2 n) is the input to system G, and Omega = Omega_1 + Omega_2, the output of system G is y[n] = |H(Omega)| cos(Omega n + Theta_H(Omega)).

Key: option D - False

8.3.a - True/false 3 - 1 points:

Question: If u[n] = e^(j Omega n) is the input to system G, and G is the cascade of two LTI systems G_1 and G_2, the output of system G is y[n] = |H_1(Omega)| |H_2(Omega)| e^(j (Omega n + Theta_H1(Omega) + Theta_H2(Omega))).

Key: option E - True

8.4.a - True/false 4 - 1 points:

Question: If u[n] = -1^n + cos(Omega n) is the input to system G, the output is y[n] = |H(Omega)| cos(Omega n + Theta_H(Omega)) + |H(0)| cos(Theta_H(0)).

Key: option H - False

Problem 9 context: Consider the continuous-time periodic signals x_i(t) = cos(w_i t + pi/4), w_i >= 0, with i = 1, 2, 3. Starting at t = 0, one period of length T_i is recorded for each signal and uniformly sampled with sampling time Ts_i: (a) T_1 = 3, Ts_1 = 1; (b) T_2 = 16, Ts_2 = 5; (c) T_3 = 16, Ts_3 = 9.

9.1.a - DFS coefficients - 7 points:

Question: Determine the DFS coefficients X_i[k] of the sampled signals for i = 1, 2, 3. Justify your answer.

Solution: x_i[n] = cos(2 pi (Ts_i/T_i) n + pi/4); the fundamental period N_i follows from Ts_i/T_i = m_i/N_i. (a) Ts_1/T_1 = 1/3 => N_1 = 3: X_1[0] = 0, X_1[1] = 3/2 e^(j pi/4), X_1[2] = 3/2 e^(-j pi/4). (b) 5/16 => N_2 = 16: only nonzero X_2[5] = 8 e^(j pi/4), X_2[11] = 8 e^(-j pi/4). (c) 9/16 => N_3 = 16: only nonzero X_3[7] = 8 e^(-j pi/4), X_3[9] = 8 e^(j pi/4). (Convention x[n] = 1/N sum_k X[k] e^(j k 2pi/N n).) Rubric: 2 points per DFS, 1 point for correct discretization.

Grading: 2 points for each DFS (3 signals); 1 point for the correct discretization.

9.2.a - Aliasing - 1 points:

Question: Determine whether aliasing occurs for i = 1, 2, 3.

Solution: No aliasing iff -pi < Omega_i < pi. x_1 (Omega_1 = 2pi/3) and x_2 (Omega_2 = 5pi/8): no aliasing. x_3 (Omega_3 = 9pi/8): aliasing.

Problem 10 context: Consider the discrete-time system G described by y[n] = 0.5 y[n-2] + u[n] - u[n-2].

10.1.a - Output DFS coefficients - 3 points:

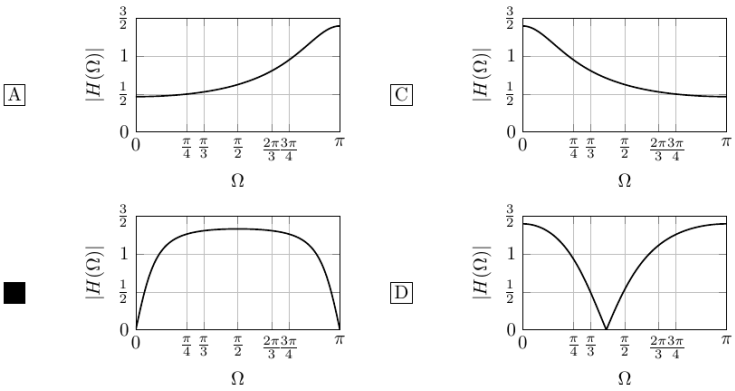
Question: If u[n] = 1 + 0.5 sin(pi/2 n) + 0.25 cos(pi n), determine the DFS coefficients of y[n]. Justify your answer.

Solution: H(Omega) = (1 - e^(-j2Omega)) / (1 - 0.5 e^(-j2Omega)). With N = 4: U[0] = 4, U[1] = -j, U[2] = 0.5, U[3] = j. Y[k] = H(e^(jk 2pi/4)) U[k] gives Y[0] = 0, Y[1] = -4/3 j, Y[2] = 0, Y[3] = 4/3 j.

10.2.a - Magnitude response plot - 2 points:

Question: Which of four plots (Plot A to Plot D, each showing a candidate magnitude response |H(Omega)| for Omega in [0, pi]) is the magnitude response of G? (see figure below) No explanation is needed.

Key: option B - Plot B

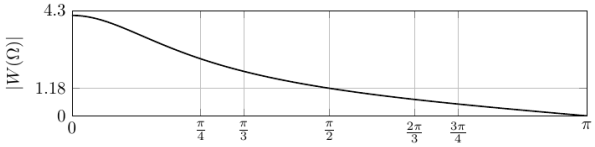


Answer options as printed; the filled box marks the correct option (B).

10.3.a - Low-pass approximation - 3 points:

Question: Consider a low-pass filter G_LP with z-transform W(z), whose magnitude response |W(Omega)| is given in a figure (value about 4.3 at Omega = 0, decreasing towards about 1.18 at Omega = pi/2) (see figure below). Which of the following approximations is true? No explanation is needed.

Key: option H - H(z) ~ W(z) W(-z)



Given magnitude response |W(Omega)| of the low-pass filter G_LP.

Problem 11 context: Consider a vector y in C^N collecting the discrete-time sequence {x[n]}, n = 0, ..., N-1, and a vector b in C^N collecting the corresponding DFT coefficients {X[k]}, k = 0, ..., N-1. The relation y = Phi b, with Phi in C^(N x N), enables reconstruction of the whole discrete-time sequence from all DFT coefficients simultaneously, or vice versa.

11.1.a - DFT matrix entries - 2 points:

Question: Determine the entries [Phi]_il, i = 0, ..., N-1, l = 0, ..., N-1, so that the DFT transform holds.

Solution: [Phi]_il = (1/N) e^(j 2pi i l / N).

11.2.a - Linearity proof - 2 points:

Question: Show that linearity holds.

Solution: Let y_1 = Phi b_1, y_2 = Phi b_2 and b = alpha_1 b_1 + alpha_2 b_2. Then y = Phi b = alpha_1 Phi b_1 + alpha_2 Phi b_2 = alpha_1 y_1 + alpha_2 y_2.